

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics Practical

Practical – I

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Class: TYIT	Batch: 01
Date of Assignment: 24/07/2026	Date/Time of Submission: 24/07/2026

1. Introduction to Business Analytics:

a. Collect data from a real-life business scenario and perform exploratory data analysis (EDA) to gain insights into the dataset.

- Import Required Libraries

```
#Import Required Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

- Load Dataset

```
#Load dataset
df = pd.read_csv("Retail_sales.csv")
```

- Display First 5 rows

```
#Display First 5 Records
df.head()
```

Output :

	Store ID	Product ID	Date	Units Sold	Sales Revenue (USD)	Discount Percentage	Marketing Spend (USD)	Store Location	Product Category	Day of the Week	Holiday Effect
0	Spearsland	52372247	2022-01-01	9	2741.69	20	81	Tanzania	Furniture	Saturday	False
1	Spearsland	52372247	2022-01-02	7	2665.53	0	0	Mauritania	Furniture	Sunday	False
2	Spearsland	52372247	2022-01-03	1	380.79	0	0	Saint Pierre and Miquelon	Furniture	Monday	False
3	Spearsland	52372247	2022-01-04	4	1523.16	0	0	Australia	Furniture	Tuesday	False
4	Spearsland	52372247	2022-01-05	2	761.58	0	0	Swaziland	Furniture	Wednesday	False

- Display Last 5 Records

```
#Display Last 5 Records
df.tail()
```

Output :

	Store ID	Product ID	Date	Units Sold	Sales Revenue (USD)	Discount Percentage	Marketing Spend (USD)	Store Location	Product Category	Day of the Week	Holiday Effect
29995	Spearsland	50239115	2022-01-25	5	2501.15	0	100	Malawi	Clothing	Tuesday	False
29996	Spearsland	50239115	2022-01-26	3	1500.69	0	0	Sudan	Clothing	Wednesday	False
29997	Spearsland	50239115	2022-01-27	6	3001.38	0	0	South Georgia and the South Sandwich Islands	Clothing	Thursday	False
29998	Spearsland	50239115	2022-01-28	5	2501.15	0	0	Haiti	Clothing	Friday	False
29999	Spearsland	50239115	2022-01-29	3	1425.66	5	190	Mozambique	Clothing	Saturday	False

- Display Shape of Dataset

```
#Display Shape of Dataset
df.shape
```

Output :

```
(30000, 11)
```

- Display Column Names

```
#Display Column Names
df.columns
```

Output :

```
Index(['Store ID', 'Product ID', 'Date', 'Units Sold', 'Sales Revenue (USD)',
      'Discount Percentage', 'Marketing Spend (USD)', 'Store Location',
      'Product Category', 'Day of the Week', 'Holiday Effect'],
      dtype='str')
```

- Display Dataset Information

```
#Display Dataset Information
df.info()
```

Output :

```
<class 'pandas.DataFrame'>
RangeIndex: 30000 entries, 0 to 29999
Data columns (total 11 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Store ID                             30000 non-null  str
1   Product ID                           30000 non-null  int64
2   Date                                 30000 non-null  str
3   Units Sold                           30000 non-null  int64
4   Sales Revenue (USD)                  30000 non-null  float64
5   Discount Percentage                  30000 non-null  int64
6   Marketing Spend (USD)                30000 non-null  int64
7   Store Location                       30000 non-null  str
8   Product Category                     30000 non-null  str
9   Day of the Week                      30000 non-null  str
10  Holiday Effect                       30000 non-null  bool
dtypes: bool(1), float64(1), int64(4), str(5)
memory usage: 2.3 MB
```

- Display Data Types

```
#Display Data Types
df.dtypes
```

Output :

```
Store ID          str
Product ID       int64
Date             str
Units Sold       int64
Sales Revenue (USD) float64
Discount Percentage int64
Marketing Spend (USD) int64
Store Location    str
Product Category  str
Day of the Week   str
Holiday Effect    bool
dtype: object
```

- Check Missing Values

```
#Check Missing Values
df.isnull().sum()
```

Output :

```

Store ID      0
Product ID    0
Date          0
Units Sold    0
Sales Revenue (USD)  0
Discount Percentage  0
Marketing Spend (USD)  0
Store Location  0
Product Category  0
Day of the Week  0
Holiday Effect  0
dtype: int64

```

- Check Duplicate Records

```

#Check Duplicate Records
df.duplicated().sum()

```

Output :

```
np.int64(0)
```

- Display Descriptive Statistics

```

#Display Descriptive Statistics
df.describe()

```

Output :

	Product ID	Units Sold	Sales Revenue (USD)	Discount Percentage	Marketing Spend (USD)
count	3.000000e+04	30000.000000	30000.000000	30000.000000	30000.000000
mean	4.461294e+07	6.161967	2749.509593	2.973833	49.944033
std	2.779759e+07	3.323929	2568.639288	5.974530	64.401655
min	3.636541e+06	0.000000	0.000000	0.000000	0.000000
25%	2.228600e+07	4.000000	882.592500	0.000000	0.000000
50%	4.002449e+07	6.000000	1902.420000	0.000000	1.000000
75%	6.559352e+07	8.000000	3863.920000	0.000000	100.000000
max	9.628253e+07	56.000000	27165.880000	20.000000	199.000000

- Count Unique Values

```

#Count Unique Values
df.nunique()

```

Output :

```

Store ID      1
Product ID    42
Date          731
Units Sold    32
Sales Revenue (USD)  2545
Discount Percentage  5
Marketing Spend (USD)  200
Store Location  243
Product Category  4
Day of the Week  7
Holiday Effect  2
dtype: int64

```

- Display Unique Values of a Particular Column

```

#Display Unique Values of a Particular Column
df["Product Category"].unique()

```

Output :

```
<StringArray>
['Furniture', 'Electronics', 'Groceries', 'Clothing']
Length: 4, dtype: str
```

- Count Product Categories

```
#Count Product Categories
df["Product Category"].value_counts()
```

Output :

```
Product Category
Furniture      9503
Electronics    8041
Clothing       6608
Groceries      5848
Name: count, dtype: int64
```

- Count Store Locations

```
#Count Store Locations
df["Store Location"].value_counts()
```

Output :

```
Store Location
Korea                237
Congo                234
Turks and Caicos Islands  156
Guam                 153
Tanzania             151
...
Palestinian Territory  98
Haiti                 96
Dominica              95
Ireland               92
Andorra               80
Name: count, Length: 243, dtype: int64
```

- Count Holiday Effect

```
#Count Holiday Effect
df["Holiday Effect"].value_counts()
```

Output :

```
Holiday Effect
False    29836
True      164
Name: count, dtype: int64
```

- Calculate Average Sales Revenue

```
#Calculate Average Sales Revenue
df["Sales Revenue (USD)"].mean()
```

Output :

```
np.float64(2749.509592666667)
```

- Find Maximum Sales Revenue

```
#Find Maximum Sales Revenue
df["Sales Revenue (USD)"].max()
```

Output :

```
np.float64(27165.88)
```

- Find Minimum Sales Revenue

```
#Find Minimum Sales Revenue
df["Sales Revenue (USD)"].min()
```

Output :

```
np.float64(0.0)
```

- Correlation Matrix

```
#Correlation Matrix
df.corr(numeric_only=True)
```

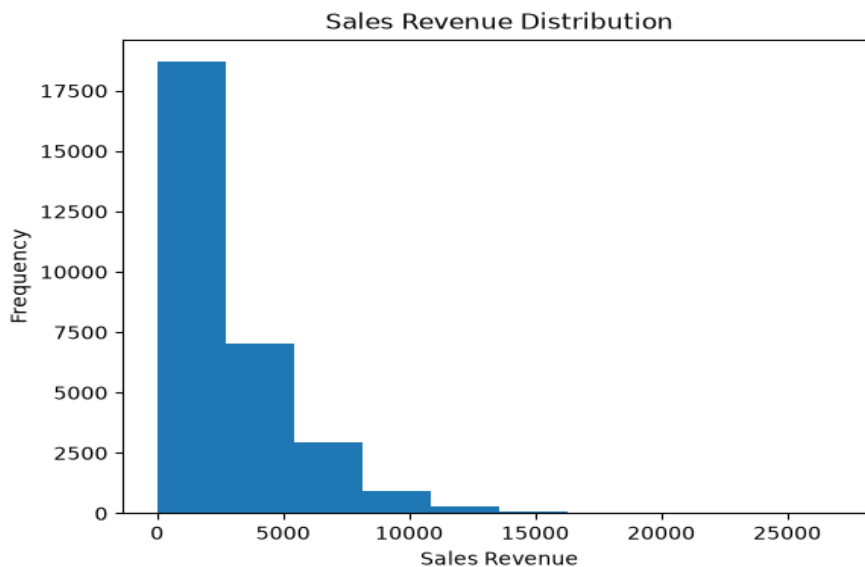
Output :

	Product ID	Units Sold	Sales Revenue (USD)	Discount Percentage	Marketing Spend (USD)	Holiday Effect
Product ID	1.000000	-0.008570	-0.181934	-0.006891	-0.002496	-0.000015
Units Sold	-0.008570	1.000000	0.615767	-0.001620	-0.004750	0.159324
Sales Revenue (USD)	-0.181934	0.615767	1.000000	-0.065791	-0.002732	0.090681
Discount Percentage	-0.006891	-0.001620	-0.065791	1.000000	-0.000322	0.005848
Marketing Spend (USD)	-0.002496	-0.004750	-0.002732	-0.000322	1.000000	-0.000455
Holiday Effect	-0.000015	0.159324	0.090681	0.005848	-0.000455	1.000000

- Histogram

```
#Histogram
plt.hist(df["Sales Revenue (USD)"])
plt.title("Sales Revenue Distribution")
plt.xlabel("Sales Revenue")
plt.ylabel("Frequency")
plt.show()
```

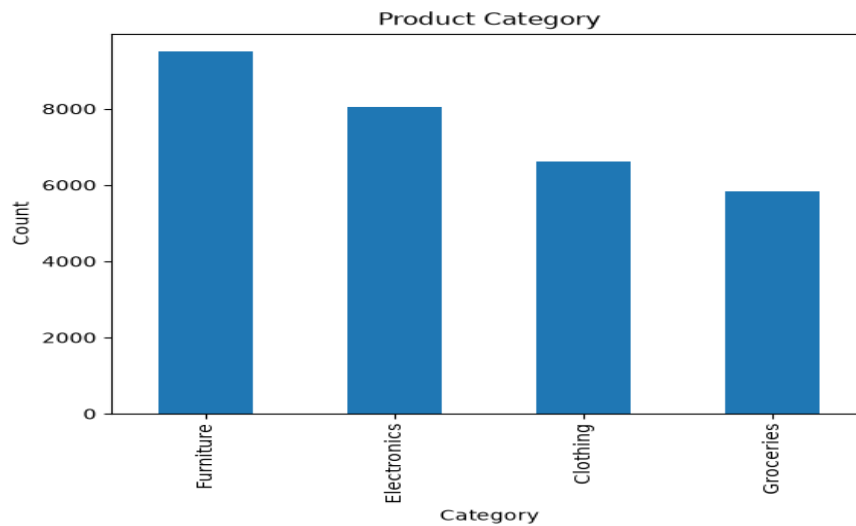
Output :



- Bar chart

```
#Bar chart
df["Product Category"].value_counts().plot(kind="bar")
plt.title("Product Category")
plt.xlabel("Category")
plt.ylabel("Count")
plt.show()
```

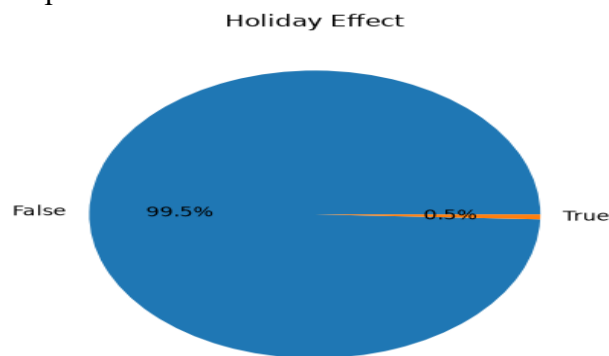
Output :



- Pie chart

```
#Pie chart
df["Holiday Effect"].value_counts().plot(kind="pie", autopct="%1.1f%%")
plt.title("Holiday Effect")
plt.ylabel("")
plt.show()
```

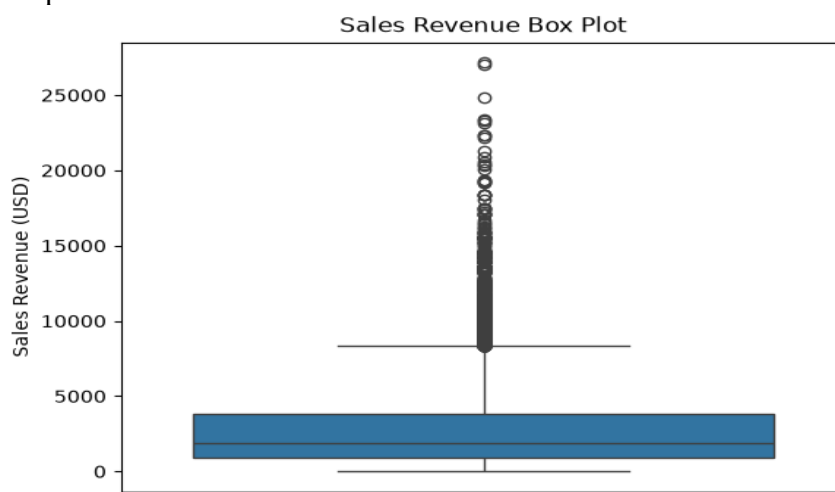
Output :



- Box plot

```
#Box plot
sns.boxplot(y=df["Sales Revenue (USD)"])
plt.title("Sales Revenue Box Plot")
plt.show()
```

Output :

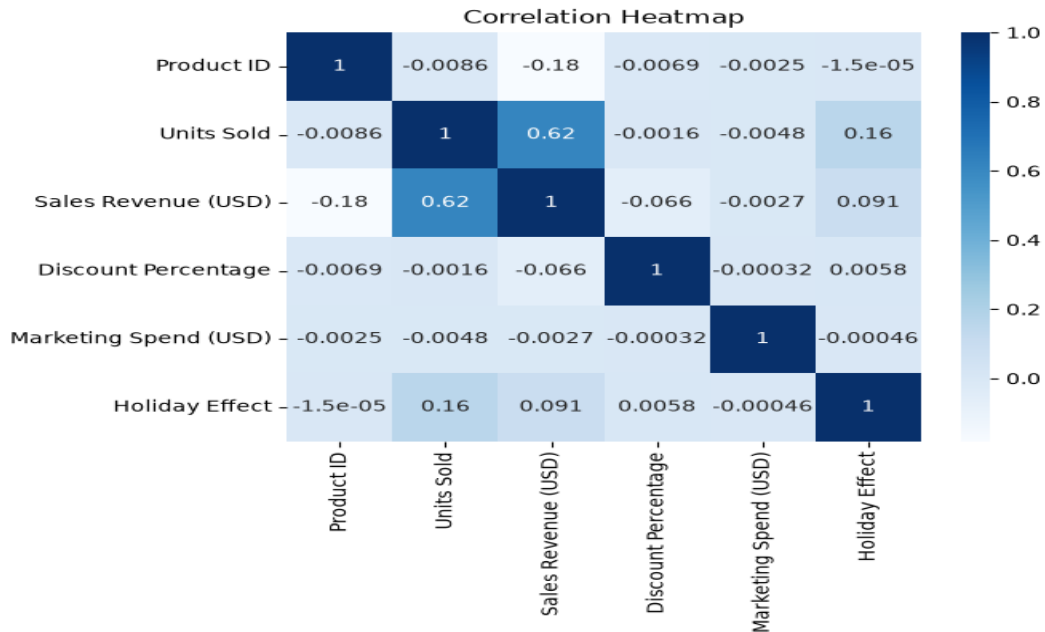


- Correlation Heatmap

```
#Correlation Heatmap
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="Blues")
```

```
plt.title("Correlation Heatmap")
plt.show()
```

Output :



b. Analyze customer data to identify trends and patterns that can be used for business decision-making.

- Import Required Libraries

```
#Import Required Libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

- Load dataset

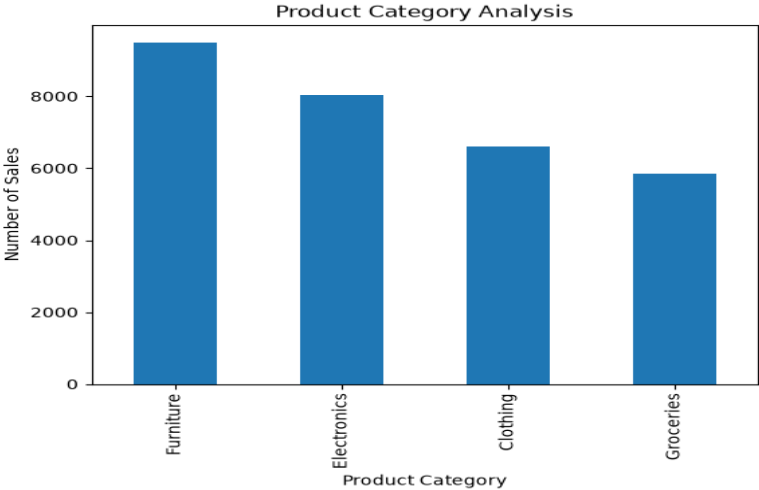
```
#Load dataset
df = pd.read_csv("Retail_sales.csv")
```

- Analysis 1 - Product category analysis

```
print(df["Product Category"].value_counts())
```

```
Product Category
Furniture      9503
Electronics    8041
Clothing       6608
Groceries      5848
Name: count, dtype: int64
```

```
df["Product Category"].value_counts().plot(kind="bar")
plt.title("Product Category Analysis")
plt.xlabel("Product Category")
plt.ylabel("Number of Sales")
plt.show()
```



● Analysis 2 - Store Location Analysis

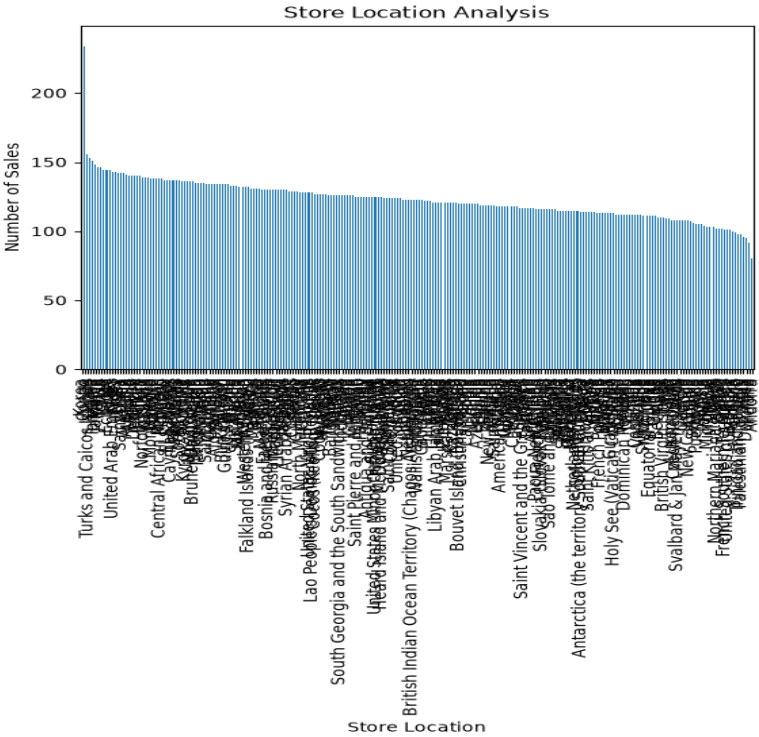
```
print(df["Store Location"].value_counts())
```

Store Location	
Korea	237
Congo	234
Turks and Caicos Islands	156
Guam	153
Tanzania	151
...	
Palestinian Territory	98
Haiti	96
Dominica	95
Ireland	92
Andorra	80

Name: count, Length: 243, dtype: int64

```
df["Store Location"].value_counts().plot(kind="bar")
plt.title("Store Location Analysis")
plt.xlabel("Store Location")
plt.ylabel("Number of Sales")
plt.show()
```

Output :



● Analysis 3 - Day of the Week Analysis

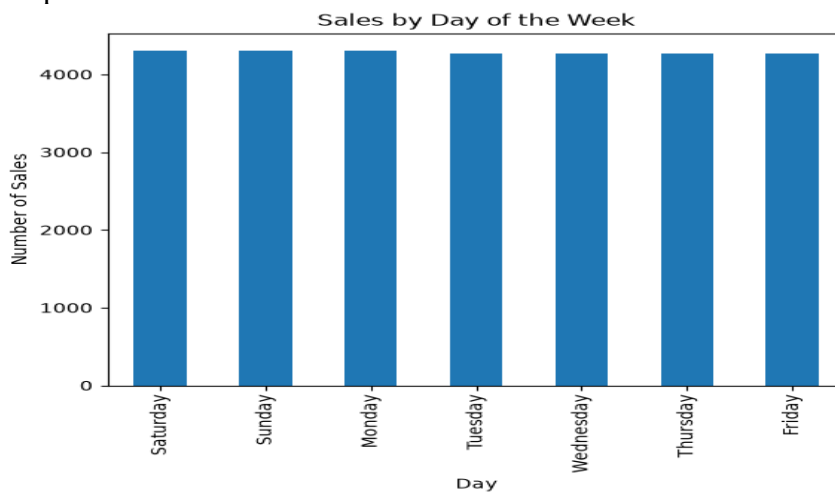
```
print(df["Day of the Week"].value_counts())
```



```
Day of the Week
Saturday      4310
Sunday        4309
Monday        4309
Tuesday       4268
Wednesday     4268
Thursday      4268
Friday        4268
Name: count, dtype: int64
```

```
df["Day of the Week"].value_counts().plot(kind="bar")
plt.title("Sales by Day of the Week")
plt.xlabel("Day")
plt.ylabel("Number of Sales")
plt.show()
```

Output :



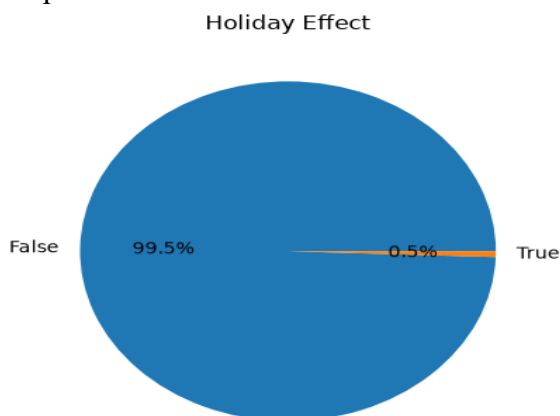
- Analysis 4 - Holiday Effect Analysis

```
print(df["Holiday Effect"].value_counts())
```

```
Holiday Effect
False      29836
True         164
Name: count, dtype: int64
```

```
df["Holiday Effect"].value_counts().plot(kind="pie", autopct="%1.1f%%")
plt.title("Holiday Effect")
plt.ylabel("")
plt.show()
```

Output :

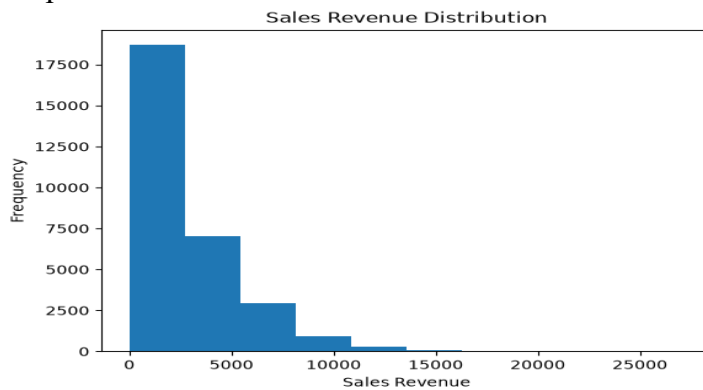


- Analysis 5 - Sales Revenue Analysis

```
plt.hist(df["Sales Revenue (USD)"], bins=10)
plt.title("Sales Revenue Distribution")
plt.xlabel("Sales Revenue")
```

```
plt.ylabel("Frequency")
plt.show()
```

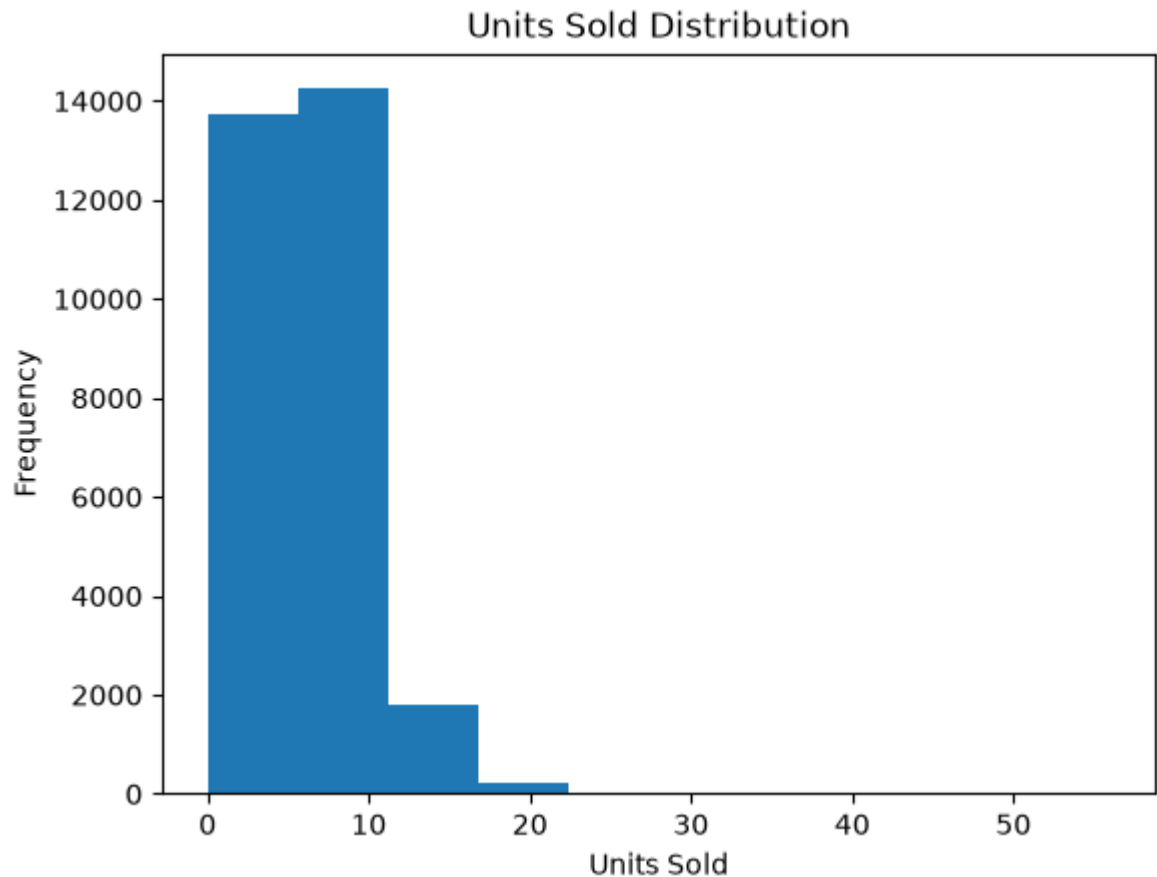
Output :



- Analysis 6 - Units Sold Analysis

```
plt.hist(df["Units Sold"], bins=10)
plt.title("Units Sold Distribution")
plt.xlabel("Units Sold")
plt.ylabel("Frequency")
plt.show()
```

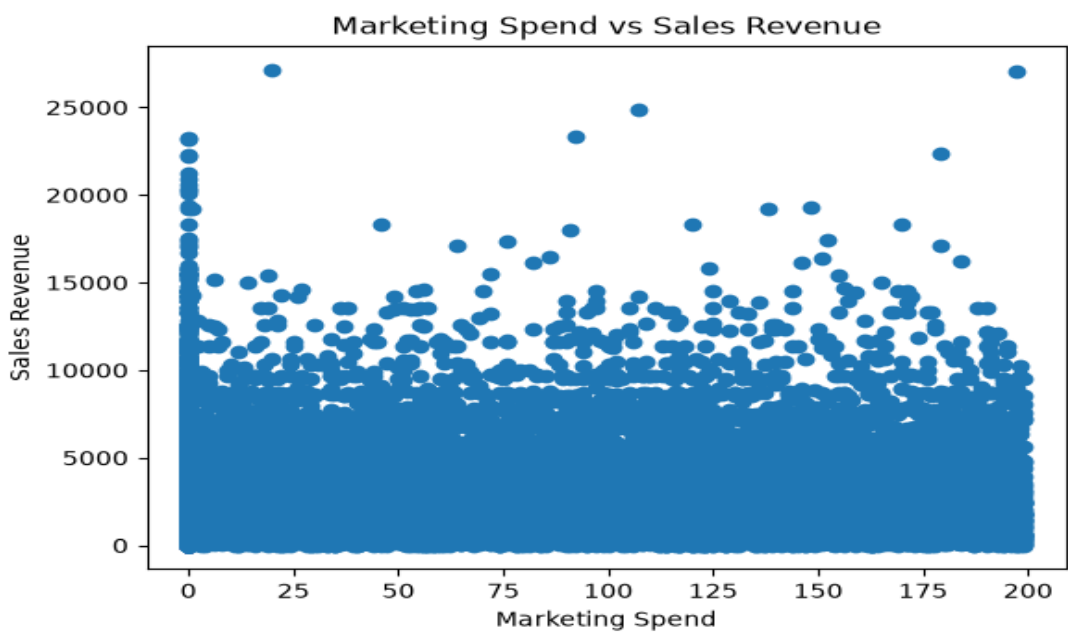
Output :



- Analysis 7 - Marketing Spend vs Sales Revenue

```
plt.scatter(df["Marketing Spend (USD)", df["Sales Revenue (USD)"])
plt.title("Marketing Spend vs Sales Revenue")
plt.xlabel("Marketing Spend")
plt.ylabel("Sales Revenue")
plt.show()
```

Output :



● Analysis 8 - Correlation Analysis

```
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="Oranges")
plt.title("Correlation Heatmap")
plt.show()
```

Output :

